

SHOW ALL YOUR WORK IN ANSWERING QUESTIONS.

We'll Done!

THINKING/INQUIRY

1. A Jaguar XK8 convertible has an eight-cylinder engine. At the beginning of its compression stroke, one of the cylinders contains 499 cm^3 of air at 1 atmospheric pressure ($1.01 \times 10^5 \text{ Pa}$) and a temperature of 27.0°C . At the end of the stroke, the air has been compressed to a volume of 46.2 cm^3 and the gauge pressure has increased to $2.72 \times 10^6 \text{ Pa}$. Calculate the final temperature. Show all your work.

[3 marks]

$$\begin{aligned} \text{Initial } 499 \text{ cm}^3 \times 10^{-6} &= 499 \times 10^{-6} \text{ m}^3 = V & p &= 1.01 \times 10^5 & T &= 300.15 \\ \text{Final } 46.2 \times 10^{-6} \text{ m}^3 &= V & p &= 2.72 \times 10^6 & & \\ \frac{P_i V_i}{T_i} &= \frac{P_f V_f}{T_f} & & & & \\ 300.15 \cdot \frac{(1.01 \times 10^5 \times 499 \times 10^{-6})}{1.01 \times 10^5 \times 499 \times 10^{-6}} &= T_f & & & & \end{aligned}$$

[3 marks]

$$\begin{aligned} T_i \frac{P_i V_i}{P_f V_f} &= T_f \\ 300.15 \times 2.85 = T_f & \\ T_f &= 776.15 \text{ K} \end{aligned}$$

Final answer: 776.15 K

2. A storage tank at STP (0°C and $1.01 \times 10^5 \text{ Pa}$) contains 28.5 kg of nitrogen gas, N_2 . The atomic mass of nitrogen is 14.01 u.

[5 marks]

- a) What is the volume of the tank? Show all your work.

$$\begin{aligned} PV &= nRT \\ V &= \frac{nRT}{P} \\ n &= \frac{28500 \text{ g}}{2 \times 14.01} \text{ mol} \\ n &= 1017.1306 \text{ mol} \\ V &= \frac{1017.1306 (8.31) (273.15)}{1.01 \times 10^5} \\ V &= 22.859 \text{ m}^3 \end{aligned}$$

Final answer: 22.86 m³

- b) What is the pressure if an additional 32.2 kg of nitrogen is added without changing the temperature? Show all your work.

$$\begin{aligned} PV &= nRT \\ P &= \frac{nRT}{V} \\ P &= \frac{(216630977) (8.31) (273.15)}{22.86 \text{ m}^3} \\ P &= 215112 \text{ Pa} \end{aligned}$$

Final answer: 215112 Pa

Name: Matias Rivas

Date: _____

3. A helium-filled balloon escapes a child's hand at sea level and 20.0°C . When it reaches an altitude of 3600 m, where the temperature is 5.0°C and the pressure only 0.68 atm, how will its volume compare to that at sea level? Consider that $1 \text{ atm} = 1.01 \times 10^5 \text{ Pa}$. Show all your work.

$$T_I = 20 + 273.15 \quad P_I = 1.01 \times 10^5 \text{ Pa}$$

$$T_F = 5 + 273.15 \quad P_F = 0.68 \times 1.01 \times 10^5 \text{ Pa}$$

$$\frac{P_I V_I}{T_I} = \frac{P_F V_F}{T_F}$$

$$V_I \left(\frac{1.01 \times 10^5 \text{ Pa}}{0.68 \times 1.01 \times 10^5 \text{ Pa}} \right) \left(\frac{273.15}{293.15} \right) = V_F$$

$$V_I \left(\frac{P_I}{P_F} \right) \left(\frac{T_F}{T_I} \right) = V_F$$

$$V_I \cdot 1.395 = V_F$$

Final answer: 1.395 times greater

[2 marks]

4. Name four different types of thermodynamic processes from the PV diagram for a constant amount of an ideal gas.

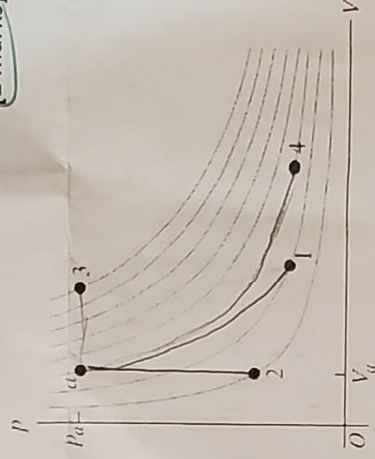
a) Adiabatic process follows the path from a to 1

b) Isochoric process follows the path from a to 2

c) Isoobaric process follows the path from a to 3

d) Isothermal process follows the path from a to 4

[2 marks]



5. Five moles of an ideal monatomic gas with an initial temperature of 127°C expand and, in the process, absorb 1500 J of heat and do 2100 J of work. What is the final temperature of the gas?

$$n = 5 \quad U = Q - W \quad U = \frac{3}{2} n R \Delta T$$

[3 marks]

$$U = 1500 - 2100$$

$$U = -600 \text{ J}$$

$$\frac{3}{2} n R \Delta T = -600 \text{ J}$$

$$\Delta T = \frac{-600 \text{ J}}{\frac{3}{2} n R}$$

$$\Delta T = -9.6295 \text{ K}$$

Final answer: 117 373°C

Name: _____

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6. When a thermodynamic system changes from an initial state to a final state, it passes through a series of intermediate states or paths.

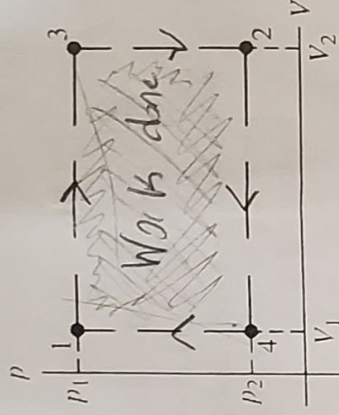
Consider a fixed mass of an ideal gas enclosed in a container fitted with a movable piston. The PV graph of this ideal gas system shows a closed loop $1 \rightarrow 3 \rightarrow 2 \rightarrow 4 \rightarrow 1$. This a cyclic process illustrates that initial and final states are the same.

[5 marks]

- a) Determine the total work done by the system, W_{total} , in this closed loop, $1 \rightarrow 3 \rightarrow 2 \rightarrow 4 \rightarrow 1$, cyclic process expressed in terms of p_1 , p_2 , V_1 and V_2 . Show all your work.

$$W = p_1(V_2 - V_1) + p_2(V_1 - V_2)$$

$$W = (p_1 - p_2)(V_2 - V_1) = \text{Area inside cycle}$$

Final answer: $(p_1 - p_2)(V_2 - V_1)$

- b) How is the work done for the process in part (a) related to the work done if the loop is traversed in the opposite direction, $1 \rightarrow 4 \rightarrow 2 \rightarrow 3 \rightarrow 1$? Explain.

The work will be negative of the work done in the initial direction. Now $3 \rightarrow 1$ is negative work and $4 \rightarrow 2$ is positive

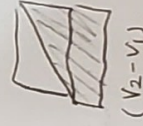
$$\text{so } p_1(V_1 - V_2) + p_2(V_2 - V_1)$$

$$(-p_1 + p_2)(V_2 - V_1)$$

$$-(p_1 - p_2)(V_2 - V_1)$$

- c) Determine the total work done by the system, W_{total} , in a path from 4 $\rightarrow$ 3 expressed in terms of p_1 , p_2 , V_1 and V_2 . Show all your work.

$$\text{Area under} = \text{Work done}$$



$$W = \frac{(p_1 - p_2)(V_2 - V_1)}{2} + p_2(V_2 - V_1)$$

$$= \frac{(p_1 - p_2)(V_2 - V_1) + 2p_2(V_2 - V_1)}{2}$$

$$= \frac{(p_1 + p_2)(V_2 - V_1)}{2}$$

Final answer: $-W_{\text{clockwise}}$

$$W_{\text{clockwise}} = -W_{\text{counterclockwise}}$$
Final answer: $\frac{(p_1 + p_2)(V_2 - V_1)}{2}$

Name: _____

Date: _____

COMMUNICATION

1. Can the temperature of a system remain constant even though heat flows into or out of it? If so, give examples.

Yes this is possible. We know that $\Delta U = Q - W$

[2 marks]

If some heat Q flows in, we can do work that is equal to Q to make $\Delta U = 0$, if $\Delta U = 0$ then there is 0 change in temperature. An example could be breathing up a gas that can do work under a piston, or doing work on a gas, but allowing the same amount of energy to flow out at the same time such as isothermal expansion for compression.

2. Name and describe four distinct types of thermodynamics processes when the amount of moles is constant. Make sure to specify how the first law of thermodynamics equation is written after simplification for each of Carnot cycle, the four types.

[6 marks]

constant temperature

Isothermal $\Delta T = 0$ so $\Delta U = 0$
 $0 = Q - W$

Isochoric $\Delta P = 0$ so $W = P\Delta V$
 $\Delta U = Q - W$

Isochoric $\Delta V = 0$

$W = 0$

Adiabatic $\Delta U = -W$

$Q = 0$

no heat flow/exchange

Isothermal means $\Delta T = 0$, adiabatic is no heat flow, isochoric is constant volume, so $W = 0$ and Isochoric is constant pressure

ΔU	Q	W	Equation
0	RT ln(V _f /V _i)	RT ln(V _f /V _i)	$0 = Q - W$
$\frac{3}{2} n R \Delta T$ $\Delta U = Q - W$	$\Delta U + W$	$P\Delta V$ $Q - \Delta U = W$	$\Delta U = Q - W$
$\frac{3}{2} n R \Delta T$ $\Delta U = Q$	$\Delta U = Q$	0	$\Delta U = Q - 0$
$\frac{3}{2} n R \Delta T$ $\Delta U = -W$	0	$\Delta U = -W$	$\Delta U = 0 - W$

Revised
N.V. 6/23

Communication marks are awarded throughout the test according to the shown rubric.

[12 marks]

	Level 1	Level 2	Level 3	Level 4
Correct use of mathematical/physical symbols, labels, and conventions (e.g. units, proper component form for the values, positive direction given, etc...)	Few mathematical/physical conventions are used correctly. [1/2]	Some mathematical/physical conventions are used correctly. [1]	Most mathematical/physical conventions are used correctly. [1-1/2]	Consistently uses mathematical/physical conventions correctly. [2]

APPLICATION

1. A scuba tank is filled with air to a gauge pressure of 204 atm when the air temperature is 29°C. A diver then jumps into the ocean and, after a short time on the ocean surface, checks the tank's gauge pressure and finds that it is only 191 atm. Assuming the diver has inhaled a negligible amount of air from the tank, what is the temperature of the ocean water? Show all your work.

$P = \text{constant}$ $PV = \frac{PV}{T} = \frac{PV}{T}$ Volume remains constant

$\frac{P_i}{T_i} = \frac{P_f}{T_f}$ Use the following

$\frac{204 \text{ atm}}{302 \text{ K}} = \frac{191 \text{ atm}}{T_f}$

$\frac{302}{204} \cdot 191 = T_f$

$T_f = 282 \text{ K}$

Final answer: 282 K

[2 marks]

2. A tire is filled with air at 15°C to a gauge pressure of 230 kPa. If the tire reaches a temperature of 38°C, what fraction of the original air must be removed if the original pressure of 230 kPa is to be maintained? Show all your work.

$T_i = 288.15 \text{ K}$ $P = 230000 \text{ Pa}$ $T_f = 311.15$

$P_i = nRT$ $PV = nRT$ $V_i = V_f$

$\frac{PV}{RT} = n$ $P = \frac{nRT}{V}$

$\frac{n_i R T_i}{V_i} = \frac{n_f R T_f}{V_f}$

$n_i T_i = n_f T_f$

$\frac{T_i}{T_f} = \frac{n_f}{n_i}$

$n_f = \frac{288.15}{311.15} n_i$

$n_i 0.92 = n_f$

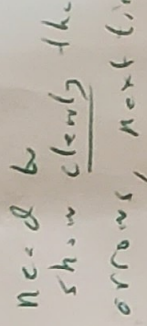
This means the new n is 0.92 initial n

[2.5 marks]

Final answer: 0.08

7.50%

- [5.5 marks]



- $$2550 \text{ T} = W$$

- $$I_{S208} = 249 \text{ (250 T)}$$

- $$(1) = 1178.350877$$

$$Q = 0 \quad \text{since isothermal}$$

$$nRT \ln\left(\frac{V_F}{V_I}\right) = nV \ln\left(\frac{V_F}{V_I}\right)$$

3025

1178.35 J